

Homotopy Theory

Summary by Ari Feiglin (ari.feiglin@gmail.com)

Contents

1	Simplicial Sets	1
1.1	Simplicial complexes	1
1.2	Simplicial sets	2
1.3	Geometric realization	3

1 Simplicial Sets

1.1 Simplicial complexes

Recall that a **simplex** in $\mathbb{R}^N$ is a subspace,

$$[v_0, \dots, v_n] = \left\{ \sum_i \lambda_i v_i \mid \sum_i \lambda_i = 1, 0 \leq \lambda_i \in \mathbb{R} \right\},$$

where $v_i \in \mathbb{R}^N$ are **geometrically independent** (if $\sum_i \lambda_i v_i = 0$ and $\sum_i \lambda_i = 0$ then $\lambda_i = 0$). The **faces** of a simplex $[v_0, \dots, v_n]$ are the simplices generated by a subset of $\{v_0, \dots, v_n\}$. The **standard n -simplex** is the one generated by the standard basis vectors of $\mathbb{R}^{n+1}$:

$$|\Delta^n| = [e_0, \dots, e_n] = \left\{ (x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid \sum_i x_i = 1, x_i \geq 0 \right\} \subseteq \mathbb{R}^{n+1}.$$

A **simplicial complex** in $\mathbb{R}^N$ is a collection of simplices K such that the face of any simplex in K is also in K , and the intersection of any two simplices in K is a face of both of them. Given a simplicial complex K , we can define its **geometric realization** $|K|$ to be the union of all the simplices in K . A subset $F \subseteq |K|$ is closed iff $F \cap \sigma$ is closed in the simplex σ for all $\sigma \in K$.

Simplicial complexes are a good starting point, but they are insufficient for our purposes. A straightforward first generalization is as follows.

1.1.1 Definition (Abstract simplicial complexes)

An **abstract simplicial complex** is a collection of sets $K = \{X^n\}_{n < \omega}$, where X^k is a collection of subsets of X^0 of cardinality $k + 1$. Furthermore, it must satisfy the condition that any subset of X^k of size $j + 1$ belongs to X^j .

Elements of X^0 are called the **vertices** of the complex.

Alternatively an abstract simplicial complex is a set K of finite subsets of X^0 which is closed under taking subsets. Then X^n is just the collection of sets in K of cardinality $n + 1$.

In a simplicial complex of $\mathbb{R}^N$ we required also that the intersection of simplices be a simplex, but this is because we were dealing with simplices embedded in an ambient space.

The standard n -simplex $|\Delta^n|$ gives rise to the abstract simplicial complex of non-empty subsets of $\{0, \dots, n\}$.

1.1.2 Definition (Face maps)

We will assume that an abstract simplicial complex K comes equipped with a total ordering on its vertices. Then for an n -simplex $\{v_0, \dots, v_n\} \in K$ we order its vertices and write $[v_0, \dots, v_n]$ to mean $v_0 < \dots < v_n$. We further define the **face maps** of the simplicial complex K to be maps $d_0, \dots, d_n: X^n \rightarrow X^{n-1}$ given by

$$d_i[v_0, \dots, v_n] = [v_0, \dots, \hat{v}_i, \dots, v_n]$$

where the hat means the vertex was removed.

For each n we have maps d_i for $i \leq n$, to differentiate the face maps between simplices of differing dimension we could write d_i^n instead of merely d_i , but this is an overload of information that is not necessary.

There is a fundamental relation between face maps which is immediately clear:

$$d_i d_j = d_{j-1} d_i, \quad \text{for } i < j. \tag{1}$$

Indeed, both sides map $[v_0, \dots, v_n]$ to $[v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_n]$.

1.1.3 Definition (Simplicial maps)

If K, L are simplicial complexes, a **simplicial map** $f: K \rightarrow L$ is a function from vertices of K to vertices of L such that if $\{v_0, \dots, v_n\} \in K$ then $\{f(v_i)\}_i \in L$. Note that a simplicial map need not be injective.

1.2 Simplicial sets

Simplicial sets, like simplicial complexes, can be given a combinatorial definition. Unfortunately the combinatorial definition is lengthier and instead has a more concise category-theoretic definition.

1.2.1 Definition (Simplicial sets)

The **simplex category** Δ is the category whose objects are finite ordinals with order-preserving maps between them. Explicitly, its objects are $[n] = \{0, \dots, n\}$ and a morphism $f: [n] \rightarrow [m]$ is a function which preserves order (not necessarily strictly).

A **simplicial set** is a functor $\Delta^{\text{op}} \rightarrow \text{Set}$. Then the category of simplicial sets is the functor category $\text{sSet} = [\Delta^{\text{op}}, \text{Set}]$. Morphisms are natural transformations.

Explicitly, a simplicial set X is a collection of sets $X_n = X([n])$ along with functions $f: X_n \rightarrow X_m$ for every order-preserving $f: [m] \rightarrow [n]$. To understand simplicial sets, it is worthwhile to understand the simplex category in more depth.

There are some obvious maps in the simplex category Δ :

- (1) The **face map** $\delta_i: [n-1] \rightarrow [n]$ (for $i \leq n$) is the unique injection whose image leaves out only i . Explicitly, $\delta_i(j) = j$ for $j < i$ and $\delta_i(j) = j+1$ for $j \geq i$.
- (2) The **degeneracy map** $\sigma_i: [n+1] \rightarrow [n]$ (for $i \leq n$) is the unique surjection which picks out i twice. Explicitly, $\sigma_i(j) = j$ for $j \leq i$ and $\sigma_i(j) = j-1$ for $j > i$.

The following identities are then easily verified:

$$\begin{aligned} \delta_i \circ \delta_j &= \delta_{j+1} \circ \delta_i \text{ for } i < j, \\ \sigma_j \circ \sigma_i &= \sigma_i \circ \sigma_{j+1} \text{ for } i \leq j, \\ \sigma_j \circ \delta_i &= \begin{cases} \delta_i \circ \sigma_{j-1} & i < j \\ \text{id} & i = j \text{ or } j+1 \\ \delta_{i-1} \circ \sigma_j & j+1 < i \end{cases}. \end{aligned}$$

Then if X is a simplicial set, we write $X(\delta_i) = d_i: X_n \rightarrow X_{n-1}$ for the image of the face maps, and $X(\sigma_j) = s_j: X_n \rightarrow X_{n+1}$ for the image of the degeneracy maps under X . These are still called the face and degeneracy maps. Note that due to the contravariance of X , the face maps “shrink” and the degeneracy maps “grow”.

The identities in the simplex category then dualize to

$$\begin{aligned} d_i d_j &= d_{j-1} d_i \text{ for } i < j, \\ s_i s_j &= s_{j+1} s_i \text{ for } i \leq j, \\ d_i s_j &= \begin{cases} s_{j-1} d_i & i < j \\ \text{id} & i = j \text{ or } j+1 \\ s_j d_{i-1} & j+1 < i \end{cases}. \end{aligned} \tag{1}$$

Note that the first identity is the same as the one in (1.1.1).

It is not hard to see that every morphism in the simplex category can be written as a composition of face and degeneracy maps. Thus we can equivalently define a simplicial set to be any collection of sets with face and degeneracy maps satisfying the identities in (1). And a morphism of simplicial sets $X \rightarrow Y$ is a sequence of functions $f_n: X_n \rightarrow Y_n$ which commute with the face and degeneracy maps: $d \circ f = f \circ d$ and $s \circ f = s \circ f$.

If K is a simplicial complex, then it defines a simplicial set X where

$$\begin{aligned} X_n &= \{[v_0, \dots, v_n] \mid \text{The } v_i \text{ form a simplex in } K \text{ with } v_0 \leq \dots \leq v_n\}, \\ d_i[v_0, \dots, v_n] &= [v_0, \dots, \hat{v}_i, \dots, v_n], \\ s_i[v_0, \dots, v_n] &= [v_0, \dots, v_i, v_i, \dots, v_n]. \end{aligned}$$

Note that the elements of X_n can be *degenerate* simplices. For example if $[0, 1, 2]$ is a simplex of K , then $[0, 0, 1, 2, 2, 2]$ is a simplex of X . Essentially the simplicial set associated with a simplicial complex is obtained by adjoining all of these degenerate simplices.

We can generalize what we mean by a degenerate simplex.

1.2.2 Definition

Let X be a simplicial set. A **degenerate simplex** of X is a simplex $x \in X$ which is in the image of some degeneracy map.

The prototypical simplicial set is

$$\Delta^n = \text{hom}_\Delta(\bullet, [n]).$$

We can better characterize Δ^n as follows. Notice that an increasing map $f: [k] \rightarrow [n]$ can be represented as a (degenerate) k -simplex $[f(0), f(1), \dots, f(k)]$. Thus we see that this definition of Δ^n coincides with the one obtained from taking the simplicial complex defined by $|\Delta^n|$ and adjoining degenerate simplices.

1.3 Geometric realization

Consider the functor $|\Delta^\bullet|: \Delta \rightarrow \mathbf{Top}$ which maps $[n]$ to $|\Delta^n|$, and an order preserving map $\varphi: [n] \rightarrow [m]$ to the continuous map $|\Delta^n| \rightarrow |\Delta^m|$ given by

$$(x_0, \dots, x_n) \in |\Delta^n| \mapsto (y_0, \dots, y_m) \in |\Delta^m|, \quad y_j = \sum_{\varphi(i)=j} x_i.$$

For example, the face maps δ_i define maps $|\Delta^{n-1}| \rightarrow |\Delta^n|$ by

$$\delta_i(x_0, \dots, x_{n-1}) = (x_0, \dots, 0, \dots, x_{n-1}),$$

and the degeneracy maps $\sigma_i: |\Delta^n| \rightarrow |\Delta^{n-1}|$ give

$$\sigma_i(x_0, \dots, x_n) = (x_0, \dots, x_i + x_{i+1}, \dots, x_n).$$

This then defines a functor

$$\text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, \bullet): \Delta^{\text{op}} \times \mathbf{Top} \rightarrow \mathbf{Set}.$$

In particular for every topological space X , we have a simplicial set

$$\mathcal{S}(X) = \text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, X),$$

this is called the **singular set** of X . The face and degeneracy maps of the singular set are then

$$d_i(\rho) = \rho \circ \delta_i, \quad s_i(\rho) = \rho \circ \sigma_i.$$

Note:

The singular set of X plays a central role in defining the singular homology of X . Recall that X 's chain complex is defined at each coordinate to be the free Abelian group generated by $\mathcal{S}(X)_n$: $C_n(X) = Z[\mathcal{S}(X)_n]$. The differentials are then defined in terms of the face maps: $\partial: C_n(X) \rightarrow C_{n-1}(X)$ is defined on a singular n -simplex $\sigma \in \mathcal{S}(X)_n$ by

$$\partial\sigma = \sum_{i=0}^n (-1)^i d_i\sigma,$$

and ∂ is extended to all of $C_n(X)$ by linearity.

Geometrically, we view a simplicial set X as a collection of (possibly degenerate) n -simplices X_n for all $n < \omega$. If we attach the geometric n -simplex $|\Delta^n|$ to each n -simplex X_n we obtain $X_n \times |\Delta^n|$. We give each X_n the discrete topology, so this becomes a topological space. The geometric realization of X as a whole should be the union of all these simplices, but this is not enough. For one, degenerate simplices should not be counted; and we want to glue the faces of simplices together.

So we define the geometric realization as follows.

1.3.1 Definition

For a simplicial set X , we define its **geometric realization** to be the space

$$|X| = \coprod_{n < \omega} X_n \times |\Delta^n| / \sim$$

where each X_n has the discrete topology and $\sim$ is the equivalence generated by the relations

$$(x, \delta_i p) \sim (d_i x, p), \quad (x, \sigma_i p) \sim (s_i x, p).$$

Here $\delta_i: |\Delta^{n-1}| \rightarrow |\Delta^n|$ and $\sigma_i: |\Delta^n| \rightarrow |\Delta^{n-1}|$ are the face and degeneracy maps respectively of the standard simplices.

For $x \in X_n$, we denote by $(x, |\Delta^n|)$ the image of $\{x\} \times |\Delta^n|$ in $|X|$.

1.3.2 Lemma

If X is a simplicial set, then every degenerate simplex of X is the degeneracy of a unique non-degenerate simplex. That is, if $x \in X$ is degenerate then there is a unique non-degenerate $y \in X$ such that $x = s_{i_1} \cdots s_{i_k} y$ for some i_j s.

Proof

Clearly if x is degenerate there must exist *some* y which satisfies this. Now suppose $x = Sy = S'y'$ for y, y' non-degenerate and S, S' a composition of degeneracy maps. Suppose $S = s_{i_1} \cdots s_{i_k}$, then define $D = d_{i_k} \cdots d_{i_1}$. This is a left inverse for S , and thus $y = DS'y'$.

Degeneracy and face maps commute according to the identities laid out in (1.2.1). Thus $y = DS'y' = \tilde{S}\tilde{D}y'$ for some composition of degeneracy maps and face maps. Since y is by assumption non-degenerate, $\tilde{S}$ must be empty so that $y = \tilde{D}y'$.

By symmetry, $y' = \bar{D}y$ for some composition of face maps as well. Since face maps lower the dimension, this can only happen if $\bar{D}, \tilde{D}$ are empty. Thus $y = y'$. ■

Similarly we have the following result.

1.3.3 Lemma

Every point $v \in |\Delta^n|$ can be expressed uniquely as $v = \delta_{i_1} \cdots \delta_{i_k} u$ where u is an interior point (all of its coordinates are positive).

Call a point $(x, v) \in |X|$ **non-degenerate** if x is non-degenerate and v is interior.

1.3.4 Lemma

For a simplicial set X , every point in its geometric realization $(x, v) \in |X|$ is equivalent to a unique non-degenerate point of $|X|$.

Proof

Define $\lambda: |X| \rightarrow |X|$ as follows. For $x \in X_n$ take $i_1, \dots, i_k$ and y non-degenerate such that $x = s_{i_1} \cdots s_{i_k} y$. Then define

$$\lambda(x, v) = (y, \sigma_{i_k} \cdots \sigma_{i_1} v).$$

Similarly define $\mu: |X| \rightarrow |X|$, which for (x, v) maps

$$\mu(x, v) = (d_{i_k} \cdots d_{i_1} x, u)$$

where $v = \delta_{i_1} \cdots \delta_{i_k} u$.

Clearly λ, μ both map points to equivalent points. Then $\lambda \circ \mu$ maps a point to an equivalent non-degenerate point.

Uniqueness is because it can be verified that if $p \sim p'$ then $\lambda\mu(p) = \lambda\mu(p')$. ■

An important fact is the following.

1.3.5 Theorem

The geometric realization of a simplicial set X is a CW complex.

Proof

Take the n -cells of $|X|$ to be the images of the non-degenerate n -simplices of X . That is, simplices $(x, |\Delta^n|)$ for $x \in X_n$ non-degenerate. By the previous lemma, these cover $|X|$, and it can be verified that these make $|X|$ into a CW complex. ■

1.3.6 Theorem

There is an adjunction $|\bullet|: \mathbf{sSet} \rightleftarrows \mathbf{Top}: \mathcal{S}$ between the geometric realization and the singular set functors.

Proof

For a simplicial set X and a topological space Y , we define maps

$$\Phi: \mathbf{hom}_{\mathbf{Top}}(|X|, Y) \rightleftarrows \mathbf{hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)): \Psi$$

as follows. For a continuous $f: |X| \rightarrow Y$, restricting f to a non-degenerate simplex $(x, |\Delta^n|)$ restricts to a continuous map $\Phi(f)_n: |\Delta^n| \rightarrow Y$.

Conversely for a map $g: X \rightarrow \mathcal{S}(Y)$, for each n -simplex $x \in X$ we have a continuous function $\rho_x: |\Delta^n| \rightarrow Y$. So define $\Psi(g): |X| \rightarrow Y$ to act on the simplex $(x, |\Delta^n|) \in |X|$ by applying ρ_x to $|\Delta^n|$. ■